

RELATIONS AND GRAPHS

Thuma E. and E. Zimudzi

Department of Computer Science
University of Botswana

30 March, 2026

OUTLINE

1 RELATIONS

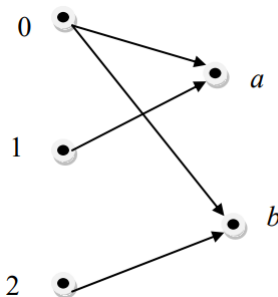
2 GRAPHS

Relations

- Let A and B be sets. A *binary relation* from A to B is a subset of $A \times B$.

A binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B . We use the notation aRb to denote that $(a, b) \in R$, i.e. a is said to be related to b by R , and $a \not R b$ to denote that $(a, b) \notin R$.

Example: Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B . Graphically, relations can be represented as:



Relations on a Set

Definition: A relation on a set A is a relation from A to A . In other words, a relation on a set A is a subset of $A \times A$.

Example 2: Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

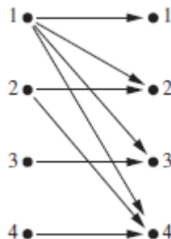


Figure: Displaying the Ordered Pairs in the Relation R from Example 2

Inverse of a Relation

Let $\mathbf{R}$ be a relation from A to B . Define the inverse relation R^{-1} from B to A as follows:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in \mathbf{R}\}$$

that is

$$\text{For all } x \in A \text{ and } y \in B, (y, x) \in R^{-1} \Leftrightarrow (x, y) \in R$$

Inverse of a Finite Relation

Example:

Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$, and let R be the “divides” relation from A to B : For every ordered pair $(x, y) \in A \times B$,

$xRy \Leftrightarrow x \mid y$. x divides y .

a. State explicitly which ordered pairs are in R and R^{-1} , and draw arrow diagrams for R and R^{-1} .

b. Describe R^{-1} in words.

Solution

$$R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$$

$$R^{-1} = \{(2, 2), (6, 2), (8, 2), (6, 3), (8, 4)\}$$

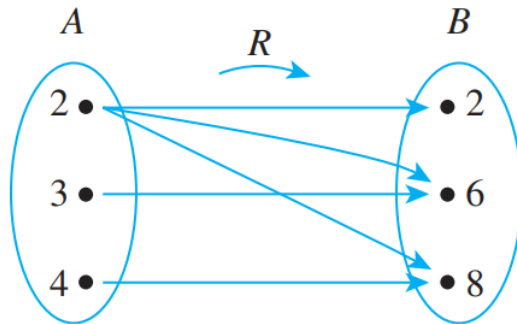


Figure: Arrow diagrams for R

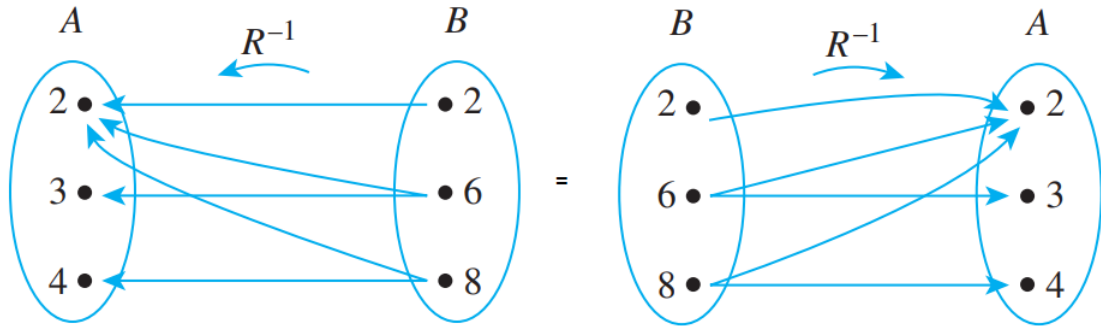


Figure: Arrow diagrams for R^{-1}

b. R^{-1} can be described in words as follows: For every ordered pair $(y, x) \in B \times A$, $yR^{-1}x \Leftrightarrow y$ is a multiple of x .

Directed Graph of a relation

Example: Let $A = \{3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows: For every $x, y \in A$, $xRy \Leftrightarrow 2 \mid (x - y)$

Solution

$3 R 3$ because $3 - 3 = 0$ and $2 \mid 0$ since $0 = 2 \cdot 0$. Thus there is a loop from 3 to itself.

Similarly, there is a loop from 4 to itself, from 5 to itself, and so forth, since the difference of each integer with itself is 0, and $2 \mid 0$.

Note also that $3R5$ because $3 - 5 = -2 = 2 \cdot (-1)$. And $5R3$ because $5 - 3 = 2 = 2 \cdot 1$. Hence there is an arrow from 3 to 5 and also an arrow from 5 to 3. The other arrows in the directed graph, as shown below, are obtained by similar reasoning.

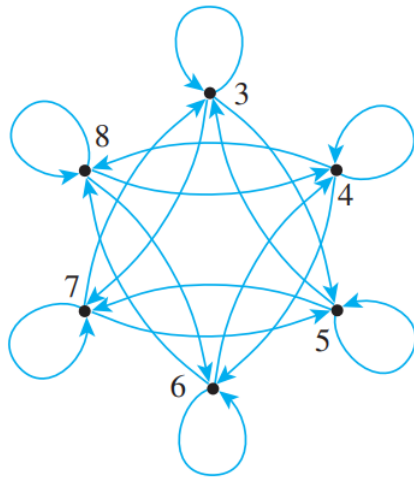


Figure: Directed Graph of a Relation

Combining Relations

Example 7: Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$.

The relations $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain:

Union $R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$

Intersection $R_1 \cap R_2 = \{(1, 1)\}$,

Difference $R_1 - R_2 = \{(2, 2), (3, 3)\}$

Difference $R_2 - R_1 = \{(1, 2), (1, 3), (1, 4)\}$

Symmetric Difference $R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2 = \{(1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$ consists of all ordered pairs which are either in R_1 or R_2 but not both.

Properties of Relations

A relation R on a set A is said to be **reflexive** if $(a, a) \in R$ for every element $a \in A$.
 $\forall a \in A, aRa$.

i.e, R is reflexive $\Leftrightarrow \forall a \text{ in } A, (a, a) \in R$

Example: $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$ is reflexive.

Definition: A relation R on a set A is called **symmetric** if $(b, a) \in R$ whenever $(a, b) \in R$.
 $\forall (a, b) \in A, aRb \Rightarrow bRa$.

i.e, R is symmetric $\Leftrightarrow \forall a$ and b in A , if $(a, b) \in R$, then $(b, a) \in R$

NB The definition of symmetric does not say x is related to y by R , rather it states that only if it happens that x is related to y , then y must be related to x .

Example: $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$ is symmetric.

Transitive A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, $\forall a, b, c \in A$.

i.e, R is transitive $\Leftrightarrow \forall a, b$ and c in A , if $(a, b) \in R$, and $(b, c) \in R$ then $(a, c) \in R$

$$aRb \text{ and } bRc \Rightarrow aRc$$

Example: $a \mid b \wedge b \mid c \Rightarrow a \mid c, \therefore \mid$ is transitive.

Example: Why is the relation below not transitive?

$$R = (1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)$$

$(3, 4) \in R$ and $(4, 1) \in R$, but $(3, 1)$ not in R .

Example

Let $A = \{0, 1, 2, 3\}$ and define relations R , S , and T on A as follows:

$$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\},$$

$$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\},$$

$$T = \{(0, 1), (2, 3)\}.$$

- a. Is R reflexive? symmetric? transitive?
- b. Is S reflexive? symmetric? transitive?
- c. Is T reflexive? symmetric? transitive?

Equivalence Relations

Let A be a set and R a relation on A . R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Example 6: Let $A = \{0, 1, 2, 3\}$

$$R_1 = \{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$$

Is R_3 an equivalence relation?

Example:

Define a relation R on $\mathbb{R}$ as follows: For all real numbers x and y , $xRy \Leftrightarrow x = y$. Is R an equivalence relation?

Solution:

a. **R is reflexive:** R is reflexive if, and only if, the following statement is true: For every $x \in \mathbb{R}$, xRx .

Since xRx just means that $x = x$, this is the same as saying

For every $x \in \mathbb{R}$, $x = x$.

But this statement is certainly true; every real number is equal to itself.

b. **R is symmetric:** R is symmetric if, and only if, the following statement is true: Define a relation R on $\mathbb{R}$ as follows: For all real numbers x and y,

For every $(x, y) \in R$, if xRy then yRx .

By definition of R, xRy means that $x = y$ and yRx means that $y = x$. Hence R is symmetric if, and only if,

For every $(x, y) \in R$, if $x = y$ then $y = x$.

But this statement is certainly true; if one number is equal to a second, then the second is equal to the first.

c. **R is transitive:** R is transitive if, and only if, the following statement is true:

For every $(x, y, z) \in R$, if xRy and yRz then xRz .

By definition of R, xRy means that $x = y$, yRz means that $y = z$, and xRz means that $x = z$. Hence R is transitive if, and only if, the following statement is true:

For every $(x, y, z) \in \mathbb{R}$, if $x = y$ and $y = z$ then $x = z$.

But this statement is certainly true: If one real number equals a second and the second equals a third, then the first equals the third.

Definition

A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E , a set of edges.

Each edge has either one or two vertices associated with it, called endpoints. The edge connects the endpoints.

Infinite graph: A graph with an infinite number of vertices.

Finite graph: A graph with a finite number of vertices.

Subgraph: A subgraph G' of $\mathbf{G}$ is a graph G' whose vertex set and edge set are subsets of those of $\mathbf{G}$

Degree of a vertex: Is the number of edges with that vertex as an endpoint. Loops give a degree of 2 to that vertex.

Parallel/Multiple edges

Example of a graph

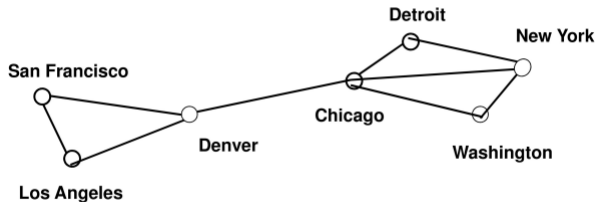


Figure: Computer Network

- Vertices, which are points in the graph, represent data centers.
- Edges, which are lines, represent communication links.
- When drawing graphs, it is advisable to avoid lines that cross each other to enhance readability of the graph, though this is not necessary.

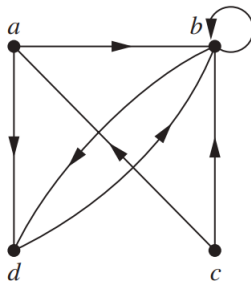
Graphs

- **Simple Graph** - Each edge connects 2 different vertices, and no two edges connect the same pair of vertices.
- **Multigraph** - The graph has multiple edges connecting the same vertices. If they are m edges associated with the same pair of vertices u, v , we say the multiplicity of the edge is m .
- **Pseudograph** - The graph has loops (edge that connects a vertex to itself).
- **Undirected Graphs** - For undirected graphs, the edges are not directed
- **Directed Graph** - also called digraph, is a graph $(G = V, E)$.
 - Where V is a nonempty set of vertices.
 - And E is a set of directed edges.
 - Each directed edge is associated with a set of ordered pairs of vertices. If a directed edge is associated with a ordered pair of vertices (u, v) , the edge is said to start at u and end at v .

A directed graph (digraph), $G = (V, E)$, consists of a non-empty set, V , of vertices (or nodes), and a set $E \subseteq V \times V$ of directed edges (or arcs). Each directed edge $(u, v) \in E$ has a start (tail) vertex u , and end (head) vertex v .

NB: a directed graph $G = (V, E)$ is simply a set V together with a binary relation E on V . An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a **loop**.

The directed graph with vertices a , b , c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is displayed below.



Example: Draw the directed graphs representing each of the relations.

a) $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\},$

b) $\{(1, 1), (1, 4), (2, 2), (3, 3), (4, 1)\},$

d) $\{(a, a), (a, b), (b, c), (c, b), (c, d), (d, a), (d, b)\}.$

Example: List the ordered pairs in the relations represented by the directed graphs.

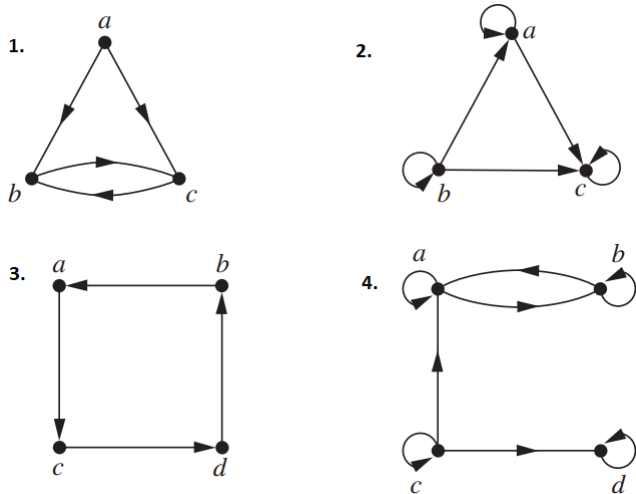


Figure: Ordered pairs in relations represented by digraphs

Example: Determine whether the relation for the directed graphs shown below is reflexive, symmetric, and/or transitive.

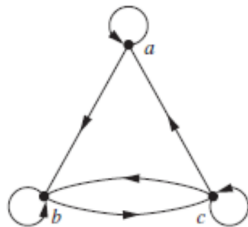
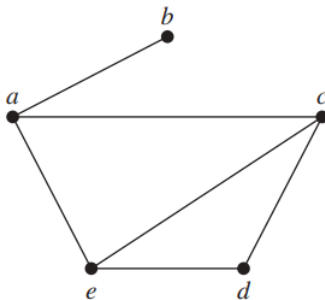


Figure: Directed graph of a relation

Sol: There are loops at every vertex of the directed graph of $\mathbf{R}$, *therefore* it is reflexive. There is an edge from a to b but not one from b to a , but there are edges in both directions connecting b and c , $\mathbf{R}$ is not symmetric. Finally, $\mathbf{R}$ is not transitive because there is an edge from a to b and an edge from b to c , but no edge from a to c .

Representing Graphs

Adjacency lists: Specify the vertices that are adjacent to each vertex of the graph. Two vertices are called adjacent if they are connected by an edge.

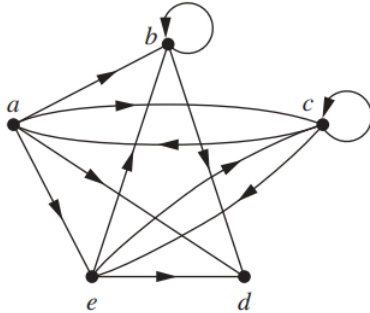


<i>Vertex</i>	<i>Adjacent Vertices</i>
<i>a</i>	<i>b, c, e</i>
<i>b</i>	<i>a</i>
<i>c</i>	<i>a, d, e</i>
<i>d</i>	<i>c, e</i>
<i>e</i>	<i>a, c, d</i>

Figure: An Adjacency list for a simple graph

Representing Graphs

Adjacency List - Directed Graphs

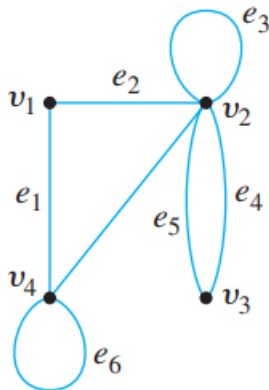


<i>Initial Vertex</i>	<i>Terminal Vertices</i>
<i>a</i>	<i>b, c, d, e</i>
<i>b</i>	<i>b, d</i>
<i>c</i>	<i>a, c, e</i>
<i>d</i>	
<i>e</i>	<i>b, c, d</i>

Figure: An Adjacency list for a Directed graph

Representing Graphs

Adjacency Matrices



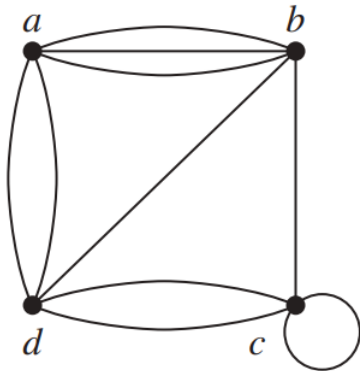
$$\mathbf{A} = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

Figure: An Adjacency Matrix of a graph

Representing Graphs

Adjacency Matrices:

Solution: The adjacency matrix using the ordering of vertices a, b, c, d is



$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

Figure: An Adjacency Matrix from the graph

Graph colouring: A colouring of a simple graph is the assignment of a colour to each vertex of the graph so that no two adjacent vertices are assigned the same colour.

Definition: The **chromatic number** of a graph is the least number of colours needed for a colouring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter chi.)

A graph G with chromatic number $\chi(G) = k$ is called k chromatic.

Application - Scheduling of University Examinations

The Department of Computer Science is offering 8 courses with final examinations at the end of the second semester of 2025/26. The table below shows (by an X) which courses have one or more students in common. Only two examination rooms are available for use at any time. Design an efficient way to schedule the final examinations.

Course Schedule								
Course	CSI132	CSI142	MAT111	MAT112	CSI131	CSI247	CSI262	CSI254
CSI132		X		X	X	X		X
CSI142	X				X	X		
MAT111						X	X	X
MAT112	X							X
CSI131	X	X				X		
CSI247	X	X	X		X		X	
CSI262			X			X		
CSI254	X		X	X				

Solution: We need to represent the potential conflicts among the schedules. A conflict exists if a course is done by 2 or more students. For instance, there is a conflict between CSI132 and CSI142 that share students. If courses do not share students, there is no conflict between them. We shall represent a conflict between two courses by drawing an edge between the corresponding vertices. The resulting graph is called a **conflict graph**.

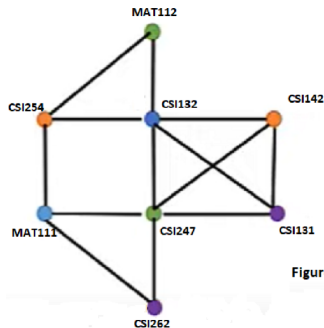


Figure: Colouring of the graph

We see that $\chi = 4$. This means 4 timeslots are needed. **Timeslot 1:** CSI132, MAT111.

Timeslot 2: MAT112, CSI247. **Timeslot 3:** CSI254, CSI142 **Timeslot 4:** CSI131, CSI262.

Practice Question: The Computer Science Department offers 10 courses next semester. Some courses cannot run at the same time. There are conflicts among the following courses

(see table):

Course	Conflicts with
A	D I
B	D I J
C	E F I
D	A B F
E	H I
F	I
G	J
H	E I J
I	A B C E F H
J	B G H

How many different time slots are needed to teach these classes (and which classes should be taught at the same time)? More importantly, how could we use graph colouring to answer this question?

The End